A Fuzzy-Embedded Edge Enhancement Network via Segment Anything Model for VHR Remote Sensing Images Change Detection

Zhiyuan Yang, Jindong Xu, *Member, IEEE*, Mengying Ni, Menghui Su, and Jiantao Peng

***Abstract*—Change detection (CD) is a fundamental yet challenging task for remote sensing (RS), with wide applications in land-cover monitoring, urban planning, disaster assessment, and environmental management. In recent years, vision foundation models (VFMs) have made breakthroughs in the field of computer vision. With their powerful zero-shot generalization ability, VFMs, represented by the segment anything model (SAM), have recently been applied to RSCD. However, existing FastSAM-based detection methods still suffer from significant shortcomings in edge detection. To address the problems of edge blurring and missed detection of small-scale changes in very-high-resolution (VHR) RS image (RSI) CD, this article proposes an edge-enhanced detection framework, FEENet, which improves FastSAM for CD on RSIs. For accurate edge detail extraction, an edge-aware module (EAM) is designed to fuse frequency-domain enhancement with semantic features. Furthermore, a multiscale fuzzy edge-guided fusion module (MFEFM) is innovatively proposed, where Gaussian fuzzy logic is employed to model edge uncertainty, and dynamic feature injection is adopted to strengthen the association between edge semantics and changed regions. Experiments on five public datasets, LEVIR-CD, WHU-CD, SYSU-CD, CLCD, and S2Looking, demonstrate that FEENet achieves better detection performance than existing methods, especially in detecting subtle changes in edge regions such as building contours and road boundaries, thus providing an effective solution for edge-aware CD tasks in VHR RSIs.**

***Index Terms*—Change detection (CD), edge enhancement, remote sensing (RS), segment anything model (SAM), vision foundation models (VFMs).**

I. Introduction

CHANGE detection (CD) plays a vital role in remote sensing image (RSI) analysis. Its core goal is to accurately identify changes between RSIs of the same area captured at different times [1]. This technology is widely used in various fields such as urban landscape monitoring [2], land-cover change [3], and disaster assessment [4], providing key data support for relevant decision-making. With the popularization of very-high-resolution (VHR) RSIs, the rich and detailed information in the images has brought unprecedented opportunities for CD but also brought severe challenges. In particular, a key challenge lies in how to fully exploit the rich information embedded in VHR imagery, which has stimulated the rapid progress of deep learning techniques in CD.

Against the backdrop of deep learning technology's rapid advancement, CD methods based on deep learning have emerged in rapid succession. Convolutional neural networks (CNNs) have been extensively applied in early CD research owing to their powerful feature extraction capabilities and can effectively learn local features in images [5], [6], [7]. However, the local convolution properties of CNNs constrain their capacity to capture long-range dependencies, hindering the comprehensive understanding of global semantic information in images. Subsequently, the emergence of Transformer brings new ideas to solve this problem. Its self-attention mechanism can model global information and show unique advantages in remote sensing CD (RSCD) tasks. For instance, BIT [8] captured spatiotemporal context through a bitemporal Transformer and achieved good results in RSCD. In addition, several recent studies have begun to explore frequency-domain representations for RSCD [9], [10], [11]. However, existing methods still have some unresolved issues when dealing with VHR RSI CD tasks, such as the lack of ability to understand the general knowledge inherent in natural images.

In recent years, vision foundation models (VFMs) have achieved significant advancements in the domain of computer vision. Among them, the segment anything model (SAM) [12] has drawn considerable interest owing to its remarkable ability to perform zero-shot segmentation. SAM is pretrained on extensive image datasets and can precisely extract semantic features across various scenarios, providing new possibilities for RSCD. As a lightweight variant of SAM, FastSAM [13] has achieved significant improvements in inference speed based on the You Only Look Once (YOLO) architecture. While maintaining certain segmentation performance, it provides possibilities for remote sensing (RS) applications with high real-time requirements. Currently, FastSAM has been introduced into the RSCD framework as a feature extractor. For example, SAM-CD [14] freezes the FastSAM encoder and designs a convolutional adapter to aggregate change-related information, taking advantage of the semantic representation capability of FastSAM and reducing dependence on vast

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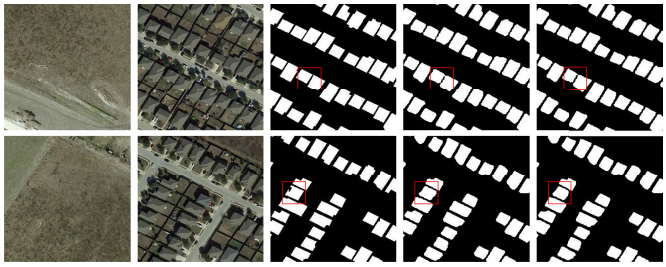


Fig. 1. Visualization of edge detection deficiencies in FastSAM-based methods.



Fig. 2. Examples of mixed pixels.

quantities of labeled data. Nonetheless, as illustrated in Fig. 1, existing CD approaches utilizing FastSAM exhibit noticeable shortcomings in edge detection precision, leading to blurred or incomplete boundaries and difficulty in reliably capturing tiny edge changes in VHR RSIs, such as changes in building outlines and the addition or disappearance of road boundaries. Directly applying SAM and its variants to VHR RSI CD still faces many challenges.

Specifically, current methods based on FastSAM mainly have two problems in edge detection. On the one hand, at the stage of feature fusion, existing FastSAM-based strategies do not explicitly model the ambiguity and uncertainty that naturally occur around object boundaries in VHR RSIs. As illustrated in Fig. 2, shadows, occlusions, and heterogeneous materials commonly lead to mixed or blurred transitional pixels along object boundaries, resulting in visually ambiguous edge regions. Without an explicit mechanism to distinguish informative high-frequency edge structures from these uncertain boundary signals, the fusion process tends to treat them as noise and suppresses them through smoothing, thereby oversmoothing the fused features in edge regions and inevitably leading to the loss of fine-grained edge details. Taking the boundary between a building and its surrounding environment as an example, in actual VHR RSIs, building edges often present a blurred transitional state due to factors such as shadows and mutual occlusion of ground objects. However, existing methods are unable to accurately depict this blurred region, leading to deviations in edge detection results. On the other hand, most methods insufficiently understand and utilize edge semantics, focusing mainly on overall semantic changes during detection and ignoring the inherent connection between edge details and change regions. This makes existing methods prone to missed detections when facing small-scale edge changes, such as the corners of newly built small buildings and residual edges after partial structure demolition.

To overcome the above problems, we propose an edge-enhanced CD framework based on FastSAM, which models edge uncertainty with fuzzy logic and employs a dynamic edge fusion module to refine edge guidance through multiscale responses. The contributions of this work are delineated as follows.

1) To address the challenges of edge blurring, boundary uncertainty, and missed detection of small-scale changes in VHR RSI CD, we propose an edge-enhanced detection framework, FEENet, built upon FastSAM.
2) An edge-aware module (EAM) is proposed to enhance low-level features in the frequency domain and integrate them with high-level semantics, thereby capturing fine structural details and suppressing background noise. A multiscale fuzzy edge-guided fusion module (MFEFM) is designed to model boundary uncertainty with Gaussian fuzzy logic and dynamically inject refined information into multiscale representations, enabling robust detection of subtle changes in complex regions.
3) Comprehensive studies conducted on five benchmark datasets, LEVIR-CD, WHU-CD, SYSU-CD, CLCD, and S2Looking, indicate that FEENet attains state-of-the-art (SOTA) performance, particularly in detecting subtle edge variations such as building contours and road boundaries.

II. Related Works

In this section, we initially examine deep learning methodologies for RSCD in Section II-A. In Section II-B, we introduce some developments in VFMs, with a special focus on SAM and its applications. Section II-C discusses the development of edge-guided CD.

A. Deep Learning-Based Methods

In recent years, deep learning-based CD methods have attained significant advancements. At the early stage of deep learning-based CD development, CNN models equipped with dual encoder structures, such as FC-EF, FC-Siam-conc, and FC-Siam-diff [5], were predominantly adopted. These fully convolutional networks employed two parallel encoders to extract features from bitemporal images. Chen et al. [15] proposed the Siamese_AUNet model, which embeds a nonlocal attention module to address the issues of weak CD and insufficient noise suppression. Fang et al. [6] modified Unet++ [16] by incorporating a Siamese architecture and employed dense skip connections to retain superficial information. DARNet [17] proposed a densely attentive refinement network to enhance CD performance for bitemporal VHR RSIs. Other research works have incorporated diverse attention modules to enhance CD performance. For instance, after the weight-sharing feature extractor, STANet [18] integrated a spatiotemporal attention module, while DMINet [19] integrated SelfAtt and CrossAtt within a unified module and incorporates an intertemporal joint-attention block to align global feature distributions between temporal inputs, thereby enhancing information exchange within each feature level and simultaneously mitigating the extraneous distractions. In addition to attention mechanisms, other CNN-based improvements such as pyramid pooling and dilated convolution have also

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been explored to enlarge the receptive field and enhance contextual modeling. Nonetheless, owing to the inherent locality of convolution operations, CNNs still struggle to capture truly global dependencies.

Transformer has gained widespread acceptance in natural language processing (NLP) owing to its robust capability to model long-range dependencies and capture global contextual information. Its architectural paradigm has since been widely adopted in computer vision, driving advances in object detection, image restoration, semantic segmentation, and image classification, among others. In the field of CD, several studies have directly employed the Transformer as the backbone network for feature extraction. For example, ChangeFormer [20] employed a hierarchical Transformer-based Siamese architecture to capture multiscale long-range dependencies, effectively surmounting the constraints of CNN-based methodologies in modeling global context for accurate CD, thereby showcasing the capabilities of Transformer in this domain. BIT captured spatiotemporal context through a bitemporal Transformer and achieved good results in RSCD. Moreover, SwinSUNet [21], constructed on the Swin Transformer block [22], integrated encoder features through a dedicated fusion module. Despite these advances, Transformer-based methods still face challenges: their limited capacity to model local details constrains overall CD performance.

To overcome this limitation, recent research has turned its focus to integrating the complementary advantages between CNNs and Transformers. In RS segmentation, it is essential to model both local contextual features and global dependencies; thus, hybrid CNN–Transformer methods [23], [24], [25], [26] have been proposed to jointly exploit the advantages of both paradigms. UNetFormer [23] proposed a U-Net-like Transformer architecture that employed CNN as the encoder and incorporated a global–local attention mechanism in the decoder to enhance segmentation performance. Similarly, CMTFNet [24] employed a CNN encoder to capture local–global contextual features. Its decoder was based on Transformer and incorporated a multiscale, multihead self-attention module. This design allowed the model to efficiently extract comprehensive multiscale global context and channel information. In addition, CTCFNet [25] designed a hybrid feature extractor composed of a multiscale pyramid vision Transformer block and a CNN block, enabling comprehensive feature representation through the complementarity of the two components. MIFNet [26] focused on alleviating cross-scale information loss by enabling more effective multiscale interaction fusion in CD networks. Apart from the above architectures, several studies have explored alternative structural paradigms for enhancing CD performance. For instance, CF-GCN [27] employed dual graph convolutions together with boundary-enhancement and knowledge-review mechanisms to strengthen spatial relational modeling, thereby improving both CD accuracy and boundary preservation in RSIs.

However, the aforementioned methods remain constrained by the traditional model training framework. Such approaches are only capable of learning the semantic information of a specific CD and fail to acquire the capability to comprehend the inherent generic knowledge of natural images. In fact, the CD task not only needs to capture the differences of bitemporal images but also relies on a general semantic understanding of ground objects such as buildings and farmland. This knowledge has been fully learned in large-scale pretrained models like CLIP [28], MAE [29], and Swin Transformer. Due to the lack of effective transfer of such general knowledge, existing methods still have the problem of insufficient robustness in small sample scenes and complex urban scenes. Therefore, how to combine the general knowledge of the pretrained model with the specific requirements of the CD task has become an important direction to improve the detection performance. Recent methods such as SAM-CD [14] and Meta-CD [30] attempt to leverage the powerful general priors of the SAM to enhance CD performance, further demonstrating the importance and necessity of integrating foundation-model knowledge into CD.

B. Vision Foundation Models

VFMs are a class of visual models that leverage large-scale pretraining to capture general visual representations. Their core value is to obtain strong general recognition ability and generalization ability through massive data learning, so as to effectively alleviate the dependence of deep learning on labeled data for downstream tasks, particularly in complex prediction challenges like semantic segmentation. Typical models of this type include CLIP [28], SAM [12], and their variants. Trained through contrastive language–image pretraining on about 400 million paired samples gathered from large-scale web data, CLIP can use text to describe visual content and achieves zero-shot classification performance approaching that of fully supervised CNNs. SAM is developed using 11 M annotated images from the SA-1B dataset, which performs zero-shot segmentation of unfamiliar objects under user-provided prompts (e.g., text or point cues), while FastSAM [13], a variant of SAM, improves inference speed by 50 times while maintaining similar generalization performance, which is more suitable for real-time scenarios. In addition, SegGPT [31], SEEM [32], and other models further expand the ability of zero-shot recognition, support multimodal prompts such as points and audio, and enhance task adaptability. However, VFM is mainly trained on natural images, which has limitations in specific domains, including RSIs, such as paying more attention to foreground objects, being difficult to deal with small objects or irregular objects, and being unable to provide semantic categories of masks. To this end, researchers optimize it by means of fine-tuning, such as designing adapters, so that it can better learn domain-specific latent semantic information for downstream tasks such as RSCD.

C. Edge-Guided Network

Edge cues are essential for enhancing the precision of salient object detection [33], [34] and other vision tasks such as CD. Usually, by introducing edge detection subnetworks, where Sobel or Canny operators are employed to generate edges, edge features are fused with nonedge features to guide the network to attend to boundary structures. As a

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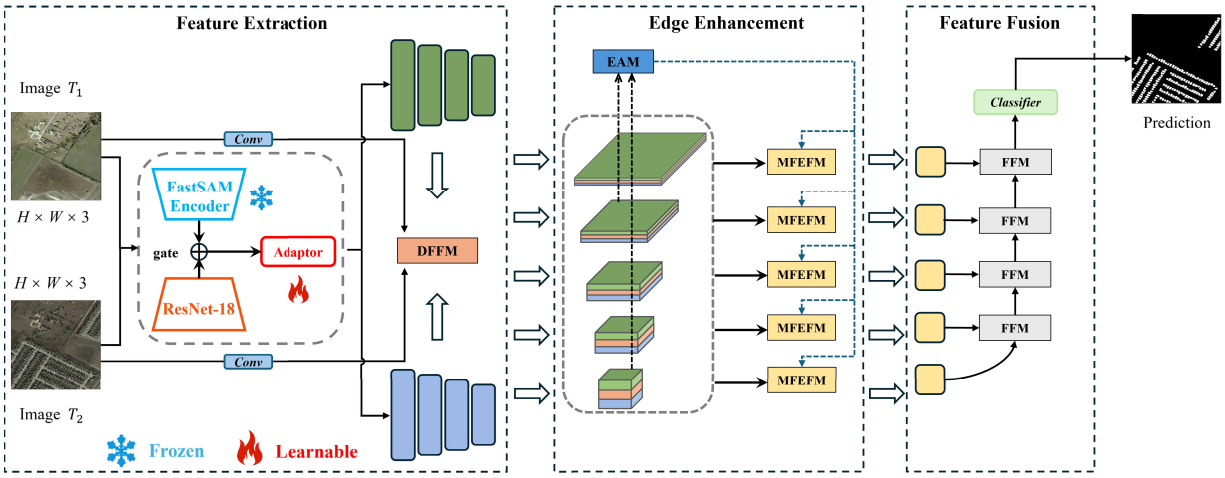


Fig. 3. Overview of FEENet.

result, clearer and more accurate edges are obtained. Among existing approaches, Cheng et al. [35] employed deformable convolutions to maximize edges, thereby clarifying the distinction between changing and invariant semantics. In addition, Bai et al. [36] developed EGRCNN, a framework that fuses discriminative information with edge structure prior knowledge to enhance CD outcomes. Meanwhile, Chen et al. [37] designed an edge-guided Transformer module for encoding the two-branch network, integrating the hierarchical Transformer with an EAM to model long-range contexts and enhance features. Moreover, Xia et al. [38] introduced an edge detection branch, utilizing edge information to restrict the output change mask. EATDer, proposed by Ma et al. [39], utilizes an edge-aware decoder. Initially, it incorporated the encoder's multiscale characteristics through a restoration block and then enhanced these features using a refining block. Finally, it generates edge detection results from the refined features and incorporates joint loss function optimization to achieve clear, smooth edges in the change map.

Inspired by [40], we design an EAM to explore the edge semantics for the specified fusion features and develop a multiscale fuzzy edge fusion module to dynamically inject the refined edge information for integration with multilevel change features, with the aim of more effectively guiding the change feature representation learning and realizing the edge-aware CD optimization.

III. Method

A. Overview

Fig. 3 outlines the comprehensive structure of the FEENet. In the encoding stage, the model employs a dual-backbone structure consisting of the FastSAM encoder and ResNet-18, which collaboratively extract discriminative semantic representations. The difference feature fusion module (DFFM) subsequently generates differential features and fuses them with the encoder outputs to enhance change-related representations. To further refine structural details, the EAM extracts precise edge cues, which are injected into multiscale feature representations through the MFEFM, thus reinforcing edge-discriminative capability. During decoding, the feature fusion module (FFM) progressively aggregates multilevel feature maps, and the fused features are input to a classifier, which yields the final CD output.

B. FastSAM Adaptor

To apply the superior feature extraction capability and generalization performance of FastSAM to the domain of RSI CD, we use its image encoder as the backbone network. The encoder is based on the YOLOv8-x model, features are extracted by CNN, and multiscale information is fused by a feature pyramid network (FPN). At the same time, ResNet-18 is introduced as a trainable collateral network, aiming to extract more discriminative semantic features in collaboration with the backbone network.

The main strategy consists in freezing the original parameters of the FastSAM encoder and only setting the ResNet-18 collateral network to a trainable state. This strategy effectively prevents interference with FastSAM pretraining parameters during the backpropagation process. In addition, we propose a trainable adapter module on the feature fusion path to adapt and align the features extracted from the backbone versus collateral networks. Suppose that t_1 and t_2 are the input bitemporal images, and $F(\cdot)$ and $R(\cdot)$ represent the FastSAM and ResNet18 feature extractors, respectively. The encoding process can be expressed as

$$F = \alpha \cdot F(t_1, t_2) + (1-\alpha) \cdot R(t_1, t_2) \tag{1}$$

$$F = \left\{ f_{t_k}^1, f_{t_k}^2, f_{t_k}^3, f_{t_k}^4 \right\}, \quad k = 1, 2 \tag{2}$$

where α is a learnable weight parameter that can automatically align the weights throughout the training phase, enabling the automatic optimization of the feature map fusion strategy during training. f is the extracted feature map and t_k represents the time. We perform feature aggregation from four different scales, denoted as $f_{t_k}^1, f_{t_k}^2, f_{t_k}^3$, and $f_{t_k}^4$.

C. Difference Feature Fusion Module

Since high-quality difference features are critical for enhancing the discriminability of subsequent change analysis tasks, extracting such information helps improve the model's

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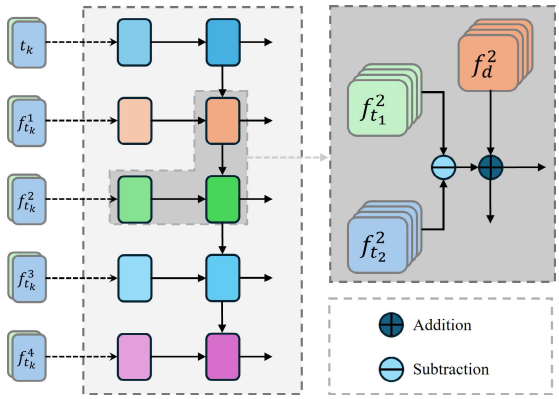


Fig. 4. Architecture of DFFM.

ability to distinguish between changed and unchanged regions. As illustrated in Fig. 4, the DFFM is designed to take two types of inputs: the raw bitemporal RSIs and the multiscale features extracted by the encoders. The raw images are first processed using a convolutional layer to adjust their channel dimension to 32. DFFM then generates informative difference features. Subsequently, these obtained difference features are employed to supplement the difference feature flow at each stage, further reinforcing the model's access to change-related information across different processing levels. The overall process can be expressed as

$$f_d^{i+1} = f_d^i + \left| f_{t_1}^i - f_{t_2}^i \right| \tag{3}$$

where f_d^{i+1} denotes the input of DFFM from stage $(i+1)$, and f_d^i denotes the output of DFFM from stage $(i-1)$. Simultaneously, $f_{t_1}^i$ and $f_{t_2}^i$ designate the outputs of the bitemporal encoder from stage i, while $|\cdot|$ is the absolute value operator.

To effectively combine the complementary information from the difference and bitemporal features, both are concatenated along the channel axis and further processed by two convolutional layers for semantic interaction. The resulting fused features enable precise localization of changes, particularly capturing fine-grained details with high accuracy. The fused features are formulated as

$$f_{\text{fused}}^i = \text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(\text{Cat}\left(f_{t_1}^i, f_{t_2}^i, f_d^i\right)\right)\right) \tag{4}$$

where $f_{t_1}^i$ and $f_{t_2}^i$ are the multiscale features of phase t_1 and phase t_2, respectively; f_d^i denotes the difference features generated by the DFFM at level i; and Cat represents the concatenation operation on the channel dimension.

D. Edge-Aware Module

EAM is designed to address the issue of inadequate accuracy in edge localization within CD tasks, especially focusing on structural information-sensitive areas such as building edges. The basic design adeptly integrates low- and high-level characteristics to accurately differentiate between change-related edges and irrelevant edges. As shown in Fig. 5, the module receives two distinct types of input features. One is the low-level feature f_{fused}^2, which contains rich edge detail information; nevertheless, it frequently includes numerous redundant edges that are not pertinent to changes. The other is the advanced feature f_{fused}^5, which has strong semantic discrimination ability, but is limited by low spatial resolution. Rather than using the first-layer feature, which is overly shallow and dominated by noisy textures, we select the second-layer feature because it strikes a better balance between rich details and stability, making it better suited to complement the semantic guidance from the fifth layer.

In the high-level feature processing branch, the channel dimension of f_{fused}^5 is first reduced by a 1×1 convolution, followed by bilinear upsampling to match the spatial resolution of f_{fused}^2. This operation ensures spatial consistency for cross-level feature fusion. The formula is as follows:

$$f_{\text{high}} = \text{Up}\left(\text{Conv}_{1\times1}\left(f_{\text{fused}}^5\right)\right) \tag{5}$$

where Up represents the bilinear upsampling operation.

To enhance low-level features, we propose a refinement mechanism based on the fast Fourier transform (FFT). Specifically, f_{fused}^2 is transformed into the frequency domain using a 2-D FFT, where an adaptive high-pass filter with a learnable cutoff frequency is applied to its amplitude spectrum. This operation adaptively preserves high-frequency components corresponding to edge information while suppressing low-frequency noise in smooth regions. The filtered amplitude spectrum is then recombined with the original phase spectrum and transformed back into the spatial domain via inverse FFT. Subsequently, the enhanced feature is scaled by a weighting factor β (set to 0.4 in our implementation) and then added to the original f_{fused}^2 through a residual connection. This operation not only preserves fine-grained spatial details but also reinforces edge-related information in a balanced manner. The operations of high-pass filtering and feature enhancement are formulated as

$$H(u,v) = \begin{cases} 1, & D(u,v) > D_0 \\ 0, & D(u,v) \le D_0 \end{cases} \tag{6}$$

$$f_{\text{low}} = f_{\text{fused}}^2 + \beta \cdot \text{IFFT}\left(F_{\text{HP}}\left(\text{FFT}\left(f_{\text{fused}}^2\right)\right)\right) \tag{7}$$

where $D(u,v)$ represents the distance from the point (u,v) to the origin, D_0 is the cutoff frequency, β is the weighting factor, and F_{HP} denotes the high-pass filtering operation.

Finally, the enhanced low-level features f_{low} and the processed high-level features f_{high} are spliced in the channel dimension, and the edge details and semantic information are deeply integrated through two layers of 3×3 convolution, and then, the edge feature map f_{edge} that highlights the change related edges is generated by 1×1 convolution to realize the accurate edge location under semantic guidance. The edge graph f_{edge} can be expressed as

$$f_{\text{edge}}^{\text{fused}} = \text{Conv}_{3\times3}\left(\text{Cat}\left(f_{\text{low}}, f_{\text{high}}\right)\right) \tag{8}$$

$$f_{\text{edge}} = \text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(f_{\text{edge}}^{\text{fused}}\right)\right) \tag{9}$$

where f_{low} denotes the enhanced low-level feature and f_{high} represents the processed high-level feature.

E. Multiscale Fuzzy Edge-Guided Fusion Module

The principal role of the MFEFM is to dynamically incorporate the edge map f_{edge} generated by the EAM into the blur-processed fused feature f_{fused}^i at each scale level. Specifically, the edge map is adaptively resized and aligned to

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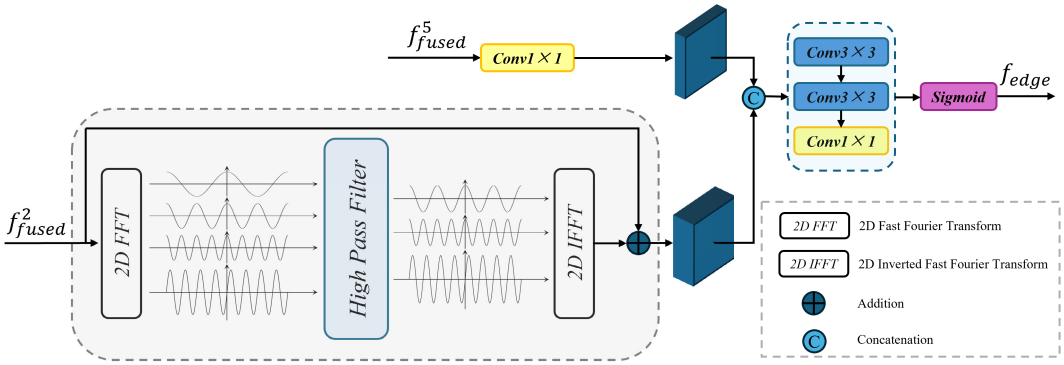


Fig. 5. Illustration of EAM.

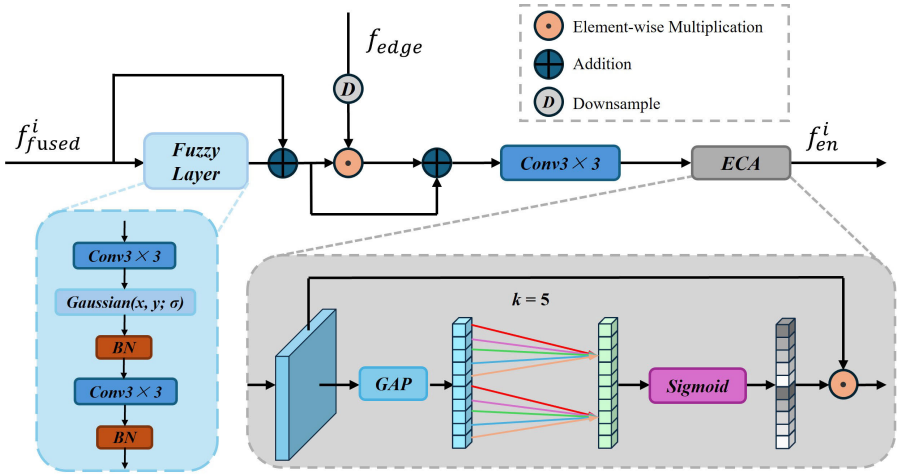


Fig. 6. Internal mechanism of the MFEFM.

match the spatial resolution of every feature layer before injection, ensuring scale-consistent guidance. After the dynamically aligned edge cues are integrated, an additional attention mechanism is applied to refine the representation, strengthening the association between edge semantics and change-related features and enhancing the model's ability to detect subtle boundary variations.

In RSIs, edge regions often exhibit significant fuzziness and uncertainty due to imaging noise, object occlusion, and mixed pixels. To effectively address this challenge, as illustrated in Fig. 6, the module first introduces a fuzzy layer based on a Gaussian membership function to process the input fusion feature f^i_{fused}. By characterizing the spatial correlations between edge pixels and their neighbors, edge uncertainty is transformed into quantifiable feature relationships, resulting in a fuzzy representation of the fusion feature.

The detailed operational process is shown in Fig. 7, which explicitly demonstrates how edge ambiguity in f^i_{fused} is modeled through spatial correlation quantification.

First, the spatial correlation weight of the pixels, denoted as w_i, in the edge region is calculated. The weight is quantified by the coordinate difference between the pixels based on the Gaussian distribution characteristics. The weight assigned to a neighboring pixel exhibits a positive correlation with its proximity to the center pixel. Specifically, the smaller the distance between the neighboring pixel and the center pixel, the greater the weight; otherwise, it will decay with

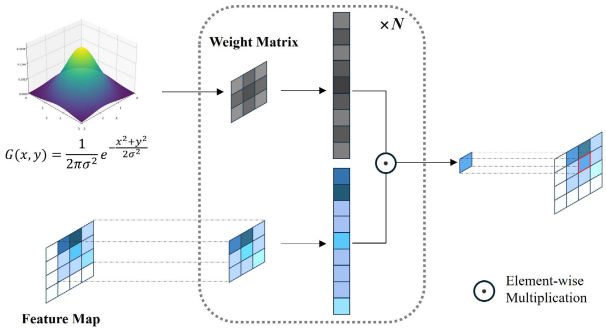


Fig. 7. Operational workflow of the fuzzy layer.

the increase of distance. The mathematical expression of this weight takes the square Euclidean distance between the center pixel coordinates (x, y) and the neighborhood pixel coordinates (x_i, y_i) as the variable, and the weight value is obtained by the exponential function mapping and is adjusted by the normalization coefficient of the Gaussian distribution. The computation process is described as follows:

$$w_i = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x_i - x)^2 + (y_i - y)^2}{2\sigma^2}\right) \tag{10}$$

where σ is a learnable parameter governing the decay rate of weights as a function of distance, thereby adjusting the fuzzy modeling range of edge features in f^i_{fused}.

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Based on the above weights, the module aggregates the feature values of neighboring pixels in f^i_{fused} to generate fuzzy edge features that incorporate spatial context information. The operation is mathematically represented as

$$F_{\text{fuzzy}} = \sum_{i=1}^{n} (w_i \cdot F_i) \tag{11}$$

where n denotes the total number of neighborhood pixels involved in the calculation and F_i denotes the feature value of the ith neighborhood pixel in f^i_{fused}. In this way, the module is able to suppress noise interference while preserving the continuity of edge details in f^i_{fused}, especially enhancing the feature representation of fuzzy transition regions.

To prevent the loss of original feature information in f^i_{fused} during fuzzy processing, the module incorporates a residual connection, directly summing the fuzzy-processed feature with the input fusion feature to get the resulting output feature

$$F'_{\text{fuzzy}} = F_{\text{fuzzy}} + f^i_{\text{fused}} \tag{12}$$

and this design makes the output feature not only contain the fuzzy edge information optimized by spatial correlation but also completely retain the original details in f^i_{fused}, which provides a feature basis with both robustness and discrimination for the subsequent edge guidance process, thereby effectively enhancing the model's capacity to differentiate fuzzy edges.

Second, to achieve spatial alignment between edge information and the fusion feature, the module applies a downsampling operation, denoted as D, to the edge map f_{edge}, so that its spatial resolution matches the current level of the fusion feature f^i_{fused}. Thereafter, the downsampled edge map interacts with the blurred fusion feature through element-wise multiplication, reinforcing the spatial correspondence between boundary structures and change regions, thus directing the model's attention to crucial boundary areas, such as changes in building outlines or their emergence.

Then, to further integrate edge semantics with multiscale change features, the module adopts a residual connection to add the element-wise multiplication result to the original blurred fusion feature. Afterward, a 3×3 convolutional layer is applied to perform deep feature fusion, generating the preliminary updated fusion feature f^i_{up}, which can be formulated as

$$f^i_{\text{up}} = \text{Conv}_{3\times3}\left(F'_{\text{fuzzy}} + D\left(f_{\text{edge}}\right) \odot f^i_{\text{fused}}\right) \tag{13}$$

where F'_{fuzzy} denotes the fuzzy processing result, $D(\cdot)$ represents the downsampling operation, and $\odot$ denotes the element-by-element multiplication. This step not only retains the basic information of the original features but also strengthens the feature interaction under the guidance of edges so that the edge details and semantic information can be effectively combined.

Finally, to enhance the discriminative power of edge-related representations, the efficient channel attention (ECA) [41] module is employed to refine f^i_{up}. Unlike conventional channel attention approaches that rely upon dimensionality reduction, ECA adopts a more lightweight yet effective strategy by directly modeling local cross-channel dependencies. Concretely, global average pooling (GAP) is initially applied to f^i_{up}, yielding the aggregated feature. Subsequently, a fast 1-D convolution with an adaptively determined kernel size of k is employed to capture local cross-channel interactions, allowing each channel to interact with its k nearest neighbors. This operation avoids the loss of channel correspondence while maintaining computational efficiency. Following this, the sigmoid function is utilized to generate the channel attention weights, and f^i_{up} is weighted and adjusted to obtain the enhanced feature f^i_{en} that can be expressed as

$$f^i_{\text{en}} = \sigma\left(F^{1D}_k\left(\text{GAP}\left(f^i_{\text{up}}\right)\right)\right) \odot f^i_{\text{up}} \tag{14}$$

where GAP$(\cdot)$ represents the global average pooling operation, F^{1D}_k denotes the 1-D convolution operation with kernel of size k, and σ represents the sigmoid function.

In summary, MFEFM balances the uncertainty and discrimination of features through the fuzzy processing mechanism, combines edge guided injection and channel attention optimization, effectively enhances the edge representation ability under multiscale, and provides key support for the improvement of edge accuracy in CD tasks.

F. Feature Fusion Module

The feature fusion module (FFM) takes the set of multiscale enhanced features f^i_{en} generated by the MFEFM as its direct input. It then employs a top-down pathway to progressively fuse them. The process begins with the deepest, semantically richest features, which are upsampled via transposed convolution to align their spatial resolution with that of the next shallower feature level. Subsequently, the upsampled deep features are fused with their corresponding shallow features, finally obtaining output features that integrate high-level semantics with low-level spatial details.

G. Loss Function

It is given as follows.

1) Throughout model optimization, BCE loss is employed to supervise pixel-level discrimination between the regions of change and no-change for CD. The loss can be expressed as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log\left(p_i\right) + \left(1 - y_i\right)\log\left(1 - p_i\right)\right]. \tag{15}$$

2) Our model incorporates a semantic learning branch to improve CD, utilizing the cosine similarity function from [14] to calculate the semantic similarity between image pairs. The resulting semantic similarity loss $\mathcal{L}_t$ is defined as

$$\mathcal{L}_t = \left[1 - \text{Cosine}\left(F_A, F_B\right)\right] \cdot (E - \text{GT}) \tag{16}$$

where F_A and F_B are the latent semantic features obtained by applying a convolutional operation to the features extracted by the encoder, GT denotes the ground truth (GT) change labels, and E is an all-ones matrix of the same size as GT. The core objective of this semantic similarity loss is to enforce feature invariance

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in the unchanged regions. To achieve this, the mask ($E - \mathrm{GT}$) is applied to exclude the changed regions from loss computation, thereby focusing the optimization on pulling the features of unchanged areas closer in the latent space.

3) Finally, the overall loss $\mathcal{L}_{\text{total}}$ is denoted as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_t. \tag{17}$$

IV. Experiments

A. Datasets

1) Levir-CD [18]: The dataset contains 637 VHR image pairs (1024 × 1024 pixels, 0.5 m/pixel) from six major Texas cities (2012–2016), capturing urban building expansion changes. It is split into 445 training, 64 validation, and 128 testing pairs and provides detailed annotations of 31 333 changed building instances.

2) WHU-CD [42]: Collected in Christchurch, New Zealand (2012–2016), the dataset spans 20.5 km^2 with a spatial resolution of 0.2 m/pixel. The original images, measuring 32 507 × 15 354 pixels, have been divided into smaller 256 × 256 pixel blocks to facilitate RSCD tasks and further split into 6096 training, 762 validation, and 762 testing pairs.

3) SYSU-CD [43]: SYSU-CD is a multicategory CD dataset spanning from 2007 to 2014, collected by Sun Yat-sen University and mainly covering the Hong Kong region. The dataset includes changes in urban housing, high-rise buildings, and coastal facilities. It consists of 20 000 RSI pairs (256 × 256 pixels, 0.5-m resolution), divided into 12 000 for training and 4000 each for validation and testing.

4) CLCD [44]: This dataset includes 600 bitemporal Gaofen-2 image pairs from Guangdong Province, China (2017–2019), with 360, 120, and 120 samples allocated for training, validation, and testing, respectively. Each 512 × 512 pair at 0.5–2-m resolution is provided with a binary change mask.

5) S2Looking [45]: S2Looking is an extensive benchmark dataset for CD, containing 5000 pairs of VHR satellite images (1024 × 1024 pixels, 0.5–0.8-m resolution). The dataset comprises about 65 920 annotated instances of building modifications and is split into training, validation, and test sets in a 7:1:2 ratio.

B. Evaluation Metrics

There are several commonly used evaluation metrics for detection tasks, including precision (P), recall (R), overall accuracy (OA), mean intersection over union (mIoU), and the mF1 score, which are defined as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{18}$$

$$\mathrm{mIoU} = \frac{1}{k} \sum_{i=0}^{k} \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{19}$$

$$\mathrm{mF1} = \frac{1}{k} \sum_{i=0}^{k} \frac{2 \times P \times R}{P + R} \tag{20}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{TN} + \mathrm{FN}} \tag{21}$$

TABLE I
QUANTITATIVE RESULTS OF VARIOUS CD METHODS ON THE LEVIR-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	mF1(%)	mIoU(%)
FC-EF	85.31	77.45	98.17	90.11	83.22
FC-Diff	90.11	80.32	98.55	92.09	86.15
FC-Conc	86.62	83.97	98.52	92.25	86.39
BIT	91.75	86.33	98.90	94.20	89.49
ChangeFormer	92.29	88.53	99.04	94.93	90.72
DMINet	91.84	88.67	99.02	94.85	90.58
CF-GCN	88.90	88.31	98.62	92.64	87.01
SAM-CD	91.66	90.59	99.10	95.32	91.37
Meta-CD	92.41	90.02	99.11	95.36	91.44
Ours	92.77	90.89	99.18	95.69	92.01

where TP and TN denote the numbers of true positive and true negative, respectively, and FP and FN represent the numbers of false positive and false negative, respectively.

C. Implementation Details

All experiments are conducted within the PyTorch framework [46] on a single NVIDIA RTX 3090 GPU. To ensure stable training, a batch size of 4 is used for the LEVIR-CD and S2Looking datasets, while a batch size of 16 is utilized for the WHU-CD, SYSU-CD, and CLCD datasets. The model is trained for 100 epochs, with the optimal checkpoint selected based on the best performance on the validation set for subsequent testing and evaluation. The proposed FEENet adopts the pretrained FastSAM-x model as its backbone. During training, the original parameters of FastSAM-x are kept frozen, and only the parameters of the supplementary components in FEENet are updated. The training process employs the SGD optimizer with an initial learning rate of 0.08, a weight decay of $5e^{-4}$, and a momentum value of 0.9.

D. Comparative Experiments

1) Comparative Methods: We selected nine methods for comparative experiments, among which FC-EF [5], FC-Conc [5], and FC-Diff [5] were implemented based on open-cd [47], and the other methods were reproduced using the official code released by the original authors. These include BIT [8], ChangeFormer [20], DMINet [19], CF-GCN [27], SAM-CD [14], and Meta-CD [30].

2) Experiments on the LEVIR-CD Dataset: Table I reports the quantitative evaluation results of different comparison methods and FEENet. The highest scores are denoted in blue, whereas the second-best results are indicated in red. FEENet regularly attains superior performance across all criteria, succeeded by Meta-CD. For the LEVIR-CD dataset, FEENet attains an OA of 99.18%, an mF1 of 95.69%, and a mIoU of 92.01%, all of which outperform the comparison methods. Compared with Meta-CD, FEENet improves mF1 by 0.27% and mIoU by 0.57%.

Fig. 8 further depicts the visual comparison of various methodologies for the LEVIR-CD dataset. FEENet demonstrates strong capability in detecting small buildings, effectively distinguishing changed regions from background interference, and achieving notably lower rates of missed and

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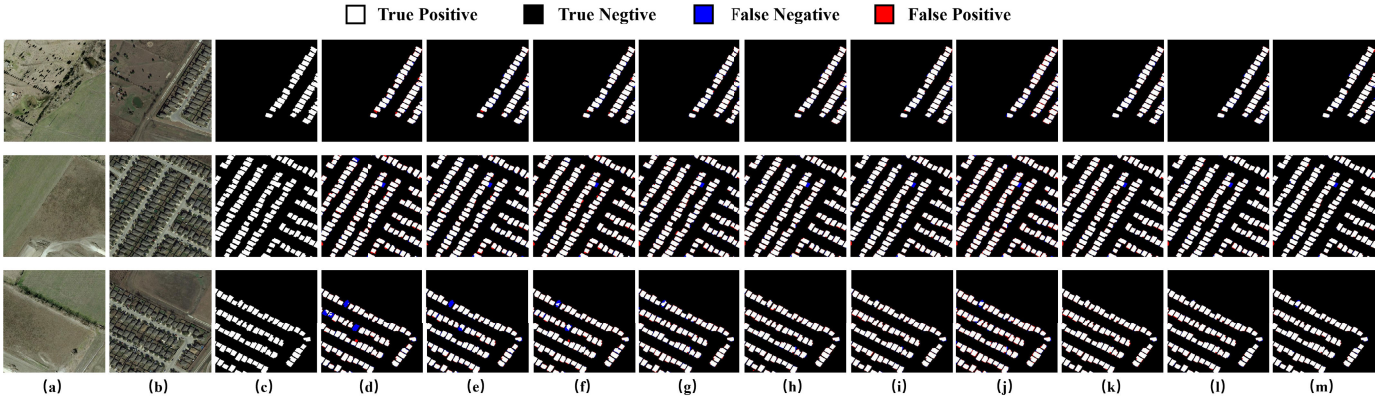


Fig. 8. Visualization on LEVIR-CD dataset. (a) T1 moment. (b) T2 moment. (c) GT. (d) FC-EF. (e) FC-diff. (f) FC-conc. (g) BIT. (h) ChangeFormer. (i) DMINet. (j) CF-GCN. (k) SAM-CD. (l) Meta-CD. (m) Ours.

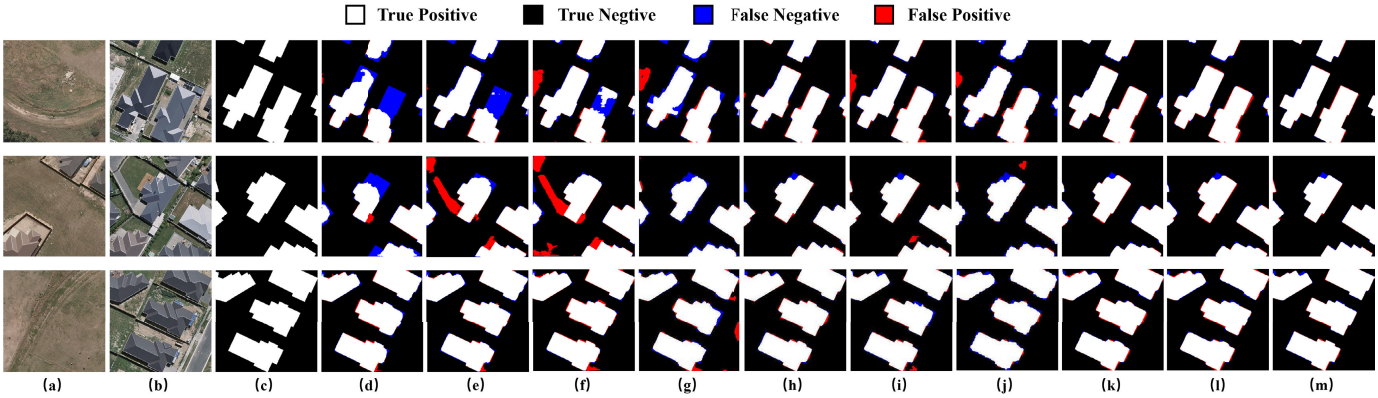


Fig. 9. Visualization on WHU-CD dataset. (a) T1 moment. (b) T2 moment. (c) GT. (d) FC-EF. (e) FC-diff. (f) FC-conc. (g) BIT. (h) ChangeFormer. (i) DMINet. (j) CF-GCN. (k) SAM-CD. (l) Meta-CD. (m) Ours.

TABLE II
QUANTITATIVE RESULTS OF VARIOUS CD METHODS ON THE WHU-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	mF1(%)	mIoU(%)
FC-EF	87.99	78.78	98.60	91.20	84.84
FC-Diff	69.85	86.04	97.96	87.97	80.21
FC-Conc	65.52	88.25	97.45	86.93	78.80
BIT	83.09	80.49	98.43	90.47	83.77
ChangeFormer	94.49	83.07	99.05	93.96	89.12
DMINet	73.07	90.12	98.11	89.86	82.84
CF-GCN	89.60	84.51	98.89	93.20	87.91
SAM-CD	94.58	90.53	99.36	96.09	92.70
Meta-CD	93.23	89.11	99.24	95.36	91.45
Ours	95.07	91.98	99.44	96.60	93.60

false detections. Moreover, compared with other approaches, FEENet excels at maintaining and restoring building edge contours. It captures contour details more precisely, preserves edge continuity, and significantly reduces aliasing and blurring, thereby providing clearer and more reliable boundaries of changed regions in VHR RS imagery. These results comprehensively verify the strong perception and reasoning ability of FEENet when handling small-scale building changes.

3) Experiments on the WHU-CD Dataset: Experiments utilizing the WHU-CD dataset are conducted to evaluate the efficacy of FEENet. The quantitative findings in Table II unequivocally illustrate the superiority of our approach in the CD task. FEENet achieves SOTA performance, with 99.44% on OA, 93.60% on mIoU, and 96.60% on mF1. Compared with SAM-CD and Meta-CD, FEENet improves mIoU by 0.90% and 2.15%, and mF1 by 0.51% and 1.24%, respectively.

The WHU-CD dataset is distinguished by its ultrahigh spatial resolution of 0.2 m, highlighting the importance of precise edge representation. FEENet effectively enhances high-frequency edge information and models fuzzy boundaries through its integrated design, enabling robust detection of subtle changes in complex regions such as building shadows and dense residential areas. The visual comparisons in Fig. 9 further confirm the advantages of FEENet in edge continuity and boundary completeness. Specifically, methods, Fig. 9(d), (e), and (f), still suffer from obvious missed detections and false detections. Although methods, Fig. 9(g)–(l), are able to detect building changes, they show poor boundary refinement, resulting in rough and discontinuous contours. In contrast, FEENet produces precise delineation of building change boundaries, demonstrating the effectiveness of our approach in enhancing model sensitivity to edge details within change regions.

4) Experiments on the SYSU-CD Dataset: Table III illustrates the quantitative experimental results of different approaches on the SYSU-CD dataset. Our method achieves superior performance on this highly challenging and diverse dataset. FEENet achieves 91.98% on OA, 88.64% on mF1,

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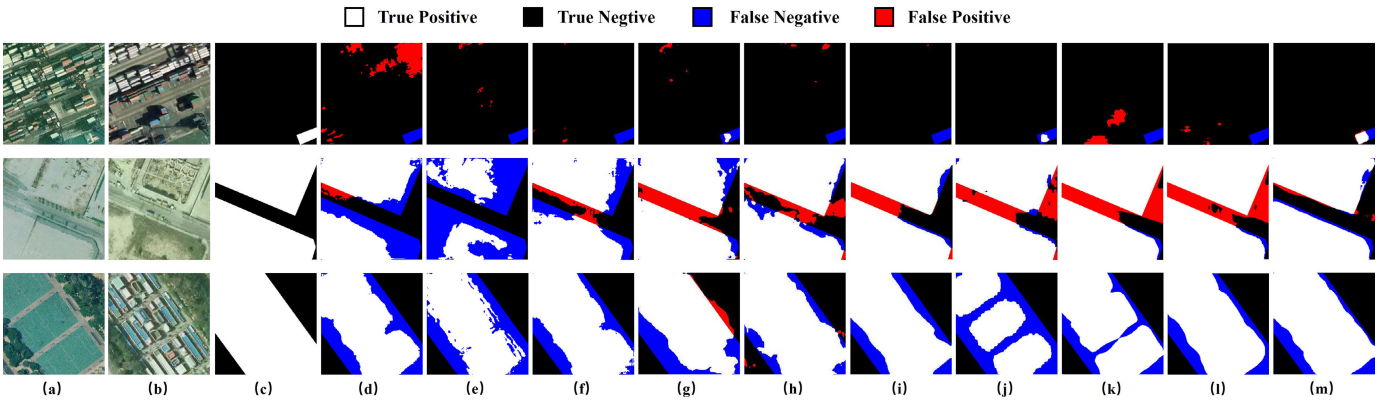


Fig. 10. Visualization on SYSU-CD dataset. (a) T1 moment. (b) T2 moment. (c) GT. (d) FC-EF. (e) FC-diff. (f) FC-conc. (g) BIT. (h) ChangeFormer. (i) DMINet. (j) CF-GCN. (k) SAM-CD. (l) Meta-CD. (m) Ours.

TABLE III
QUANTITATIVE RESULTS OF VARIOUS CD METHODS ON THE SYSU-CD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	mF1(%)	mIoU(%)
FC-EF	75.28	70.35	87.56	82.34	71.11
FC-Diff	83.49	48.32	85.56	76.17	63.90
FC-Conc	79.43	69.92	88.64	83.53	72.79
BIT	80.17	72.00	89.19	84.45	74.05
ChangeFormer	79.37	72.66	89.10	84.42	73.98
DMINet	85.08	74.06	90.82	86.65	77.21
CF-GCN	80.31	72.58	89.34	84.69	74.37
SAM-CD	78.02	82.21	90.34	86.84	77.38
Meta-CD	76.27	82.96	89.89	86.39	76.69
Ours	85.06	80.06	91.98	88.64	80.15

TABLE IV
QUANTITATIVE RESULTS OF VARIOUS CD METHODS ON THE CLCD DATASET

Methods	Pre(%)	Rec(%)	OA(%)	mF1(%)	mIoU(%)
FC-EF	73.65	55.48	95.30	80.75	71.08
FC-Diff	78.84	51.36	95.29	79.85	70.12
FC-Conc	66.48	66.49	94.95	81.88	72.24
BIT	65.68	55.67	94.69	78.71	68.80
ChangeFormer	64.05	59.70	94.67	79.47	69.57
DMINet	59.97	58.27	94.17	77.98	67.93
CF-GCN	65.01	48.66	94.40	76.34	66.38
SAM-CD	70.72	73.81	95.90	85.01	76.10
Meta-CD	74.28	77.64	96.44	87.00	78.71
Ours	74.61	77.11	96.45	86.96	78.66

and 80.15% on mIoU. Relative to the second-best result, the proposed method enhances OA, mF1, and mIoU by 1.16%, 1.80%, and 2.77%, respectively, which fully verifies its effectiveness in complex scenarios.

The visualization results of FEENet and other competing methods on the SYSU-CD dataset are displayed in Fig. 10. The results show that FEENet can successfully identify multiscale building changes in complex scenes. Specifically, in the results of line 1, except for the methods BIT and CF-GCN, the remaining deep learning methods cannot effectively detect small-scale changes. On the contrary, FEENet can precisely identify small-scale change regions with fewer missed detections and false detections. Moreover, thanks to the designed edge-aware mechanism, FEENet achieves clearer and more continuous boundaries around changed objects, effectively reducing edge blurring. In addition, the visualization results in rows 2 and 3 further show that the prediction performance of FEENet is still ahead of other comparison methods in different complex scenes, demonstrating the effectiveness of the proposed approach.

5) Experiments on the CLCD Dataset: The results of the experiments on the CLCD dataset further corroborate the efficacy of FEENet in small-sample CD scenarios. The CLCD dataset contains 600 pairs of arable land change samples, making it a relatively small-scale dataset. Meta-CD, designed specifically for few-shot CD tasks, typically enjoys a natural advantage on such data.

As shown in Table IV, despite the strong performance of Meta-CD in small-sample processing, FEENet still achieves comparable results. Specifically, FEENet attains 96.45% OA, slightly surpassing the 96.44% obtained by Meta-CD. Its mF1 reaches 86.96%, only marginally lower than the 87.00% achieved by Meta-CD, yet notably higher than the 85.01% of SAM-CD and other comparison methods. For mIoU, FEENet achieves 78.66%, less than 0.1% below the 78.71% of Meta-CD but 2.56% higher than SAM-CD, reflecting its accurate localization capability in detecting cropland changes.

Fig. 11 presents the visualization results of FEENet and other comparison methods on the CLCD dataset. In the first row, methods, Fig. 11(d)–(l), exhibit obvious false detections in the lower right region, while FEENet effectively suppresses such spurious responses, yielding more reliable predictions. In the third row, FEENet also demonstrates superior recognition and handling of elongated change regions, achieving more complete and continuous detection of linear land-cover changes. It also achieves the lowest miss and false detections in other cases, further demonstrating the superiority of our method.

6) Experiments on the S2Looking Dataset: Table V presents the quantitative comparison results on the S2Looking dataset. Our method achieves superior performance over existing approaches in terms of mF1, mIoU, and OA. Specifically, compared with the suboptimal method, it yields improvements of 0.02% in OA, 0.17% in mF1, and 0.19% in mIoU on this

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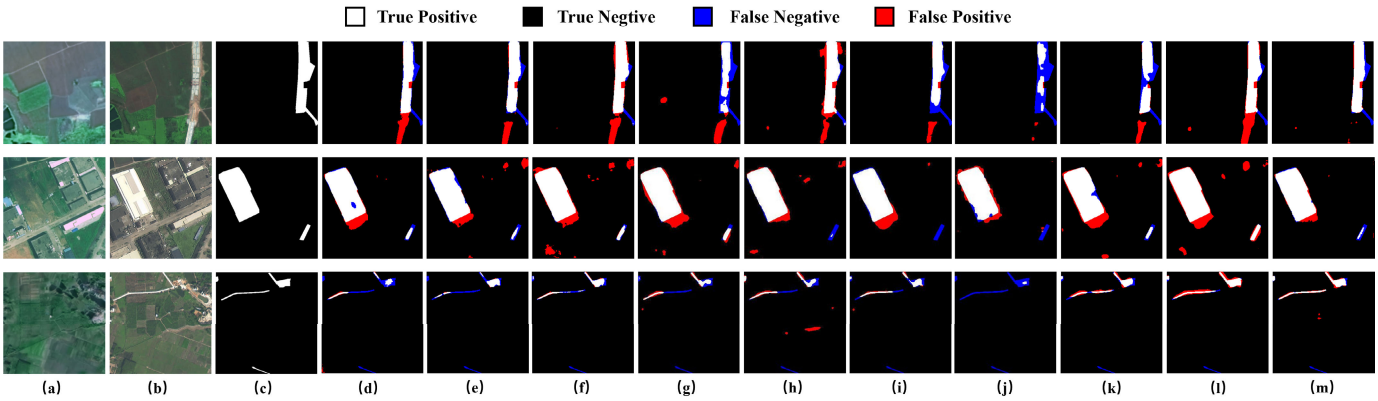


Fig. 11. Visualization on CLCD dataset. (a) T1 moment. (b) T2 moment. (c) GT. (d) FC-EF. (e) FC-diff. (f) FC-conc. (g) BIT. (h) ChangeFormer. (i) DMINet. (j) CF-GCN. (k) SAM-CD. (l) Meta-CD. (m) Ours.

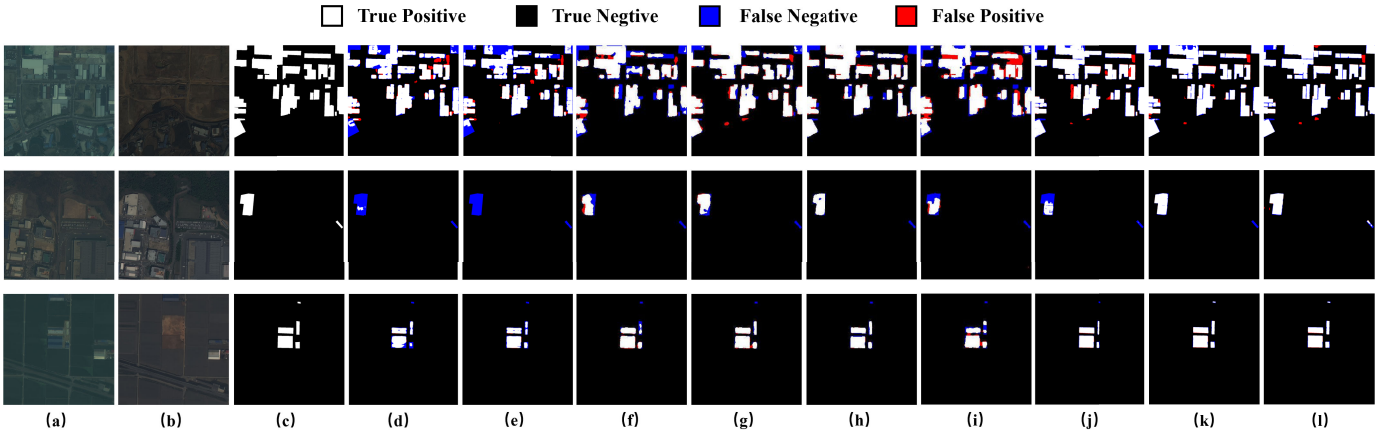


Fig. 12. Visualization on S2Looking dataset. (a) T1 moment. (b) T2 moment. (c) GT. (d) FC-diff. (e) FC-conc. (f) BIT. (g) ChangeFormer. (h) DMINet. (i) CF-GCN. (j) SAM-CD. (k) Meta-CD. (l) Ours.

TABLE V
QUANTITATIVE RESULTS OF VARIOUS CD METHODS ON THE S2LOOKING DATASET

Methods	Pre(%)	Rec(%)	OA(%)	mF1(%)	mIoU(%)
FC-EF	-	-	-	-	-
FC-Diff	73.80	32.93	99.05	72.53	64.26
FC-Conc	78.80	30.34	99.06	71.67	63.55
BIT	63.07	35.88	98.97	72.61	64.31
ChangeFormer	66.36	45.20	99.05	76.65	67.92
DMINet	69.01	44.19	99.08	76.71	67.98
CF-GCN	58.81	22.06	98.87	65.76	58.99
SAM-CD	79.39	53.10	99.23	81.13	72.42
Meta-CD	72.33	57.03	99.23	81.69	73.01
Ours	76.52	55.14	99.25	81.86	73.20

complex dataset, highlighting the robustness of our model. Furthermore, the results confirm that our method maintains strong performance even in complex change scenarios.

Visual comparison results on the S2Looking dataset are depicted in Fig. 12. As shown, the proposed method achieves superior performance: in the first row, our method produces the fewest missed and false detections; in the second row, only our method effectively identifies the change on the right; in the last row, only our method and Meta-CD are capable of detecting the little change at the top. Meanwhile, our method also provides more precise edge details. These visual comparison results further confirm the effectiveness and robustness of our method in capturing subtle changes and preserving edge details under complex scenarios.

E. Ablation Study

1) Effect of FastSAM Encoder: To clarify the impact of introducing the ResNet-18 encoder on model performance, we designed two sets of ablation experiments for comparison. The first group uses only FastSAM for feature extraction, while the second group fuses FastSAM with features extracted by ResNet-18. To provide an equitable comparison, the subsequent processing procedures are kept identical for both groups. As shown in Table VI, introducing the ResNet-18 encoder consistently outperforms using only FastSAM across multiple metrics. For the LEVIR-CD dataset, OA increases by 0.01%, mIoU by 0.04%, and mF1 by 0.02%. For the WHU-CD dataset, OA improves by 0.03%, mIoU by 0.30%, and mF1 by 0.17%. The results validate the beneficial impact of the ResNet-18 encoder on improving model performance.

From the perspective of feature representation, this performance improvement arises from the complementarity between

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TABLE VI
ABLATION STUDY ON FASTSAM ENCODER

Encoder	LEVIR-CD			WHU-CD		
	OA(%)	mIoU(%)	mF1(%)	OA(%)	mIoU(%)	mF1(%)
FastSAM	99.17	91.97	95.67	99.41	93.30	96.43
FastSAM+ResNet-18	**99.18**	**92.01**	**95.69**	**99.44**	**93.60**	**96.60**

TABLE VII
ABLATION STUDY ON DFFM, EAM, AND MFEFM

Method	Module			LEVIR-CD			WHU-CD		
	DFFM	EAM	MFEFM	OA(%)	mIoU(%)	mF1(%)	OA(%)	mIoU(%)	mF1(%)
Baseline	×	×	×	99.01	90.56	94.80	99.18	91.19	95.06
FEENet	✓	×	×	99.03	90.74	94.95	99.21	91.30	95.27
FEENet	✓	✓	×	99.12	91.47	95.38	99.35	92.64	96.06
FEENet	✓	×	✓	99.15	91.76	95.55	99.43	93.45	96.52
FEENet	✓	✓	✓	**99.18**	**92.01**	**95.69**	**99.44**	**93.60**	**96.60**

the two feature extractors. FastSAM, built upon the YOLO architecture, is adept at capturing local patterns and fine-grained textures through convolutional operations, making it effective for contour localization. However, its capability for global semantic modeling and contextual understanding in complex scenes is relatively limited. In contrast, ResNet-18 leverages residual connections to gradually aggregate multiscale features, thereby enhancing semantic representation while preserving local details, which is particularly useful for modeling spatial relationships among objects in RSIs. By combining the strengths of both extractors, the model can not only take advantage of FastSAM's efficiency in capturing local details for precise recognition of subtle boundary or corner changes but also benefit from ResNet-18's multiscale semantic representation to better characterize large-scale land-cover transformations.

Comprehensive experimental results and feature characteristics introducing ResNet-18 features can effectively make up for the shortcomings of FastSAM in global semantic modeling and improve the model's capacity to identify diverse alterations in intricate RS situations. Therefore, this study chooses the feature fusion scheme of FastSAM and ResNet-18 as the feature extractor.

2) Impact of DFFM, EAM, and MFEFM: To accurately evaluate the contribution of each component in our framework, we constructed a baseline model that uses the hybrid encoder composed of FastSAM and ResNet-18 to extract multilevel features, which are then directly decoded to generate the final CD prediction. This baseline serves as the fundamental version of our architecture.

Based on this baseline, we first incorporate the DFFM module to explicitly extract difference features between the bitemporal inputs. As shown in Table VII, introducing DFFM brings clear performance improvements on both datasets. On the LEVIR-CD dataset, the OA, mIoU, and mF1 increase by 0.02%, 0.18%, and 0.15%, respectively. On the WHU-CD dataset, these metrics improve by 0.03%, 0.11%, and 0.21%. These results demonstrate that constructing explicit difference representations enables the model to more effectively identify changes while suppressing irrelevant temporal noise, thereby enhancing the fundamental discriminative capability of the network.

Building upon the baseline equipped with DFFM, we further incorporate the EAM and MFEFM modules separately. Both modules yield substantial improvements over the baseline with DFFM. On the two datasets, adding EAM improves the OA by 0.09% and 0.14%, the mIoU by 0.73% and 1.34%, and the mF1 by 0.43% and 0.79%, confirming its effectiveness in extracting fine-grained edge information. Similarly, introducing MFEFM brings consistent gains, increasing the OA by 0.12% and 0.22%, the mIoU by 1.02% and 2.15%, and the mF1 by 0.60% and 1.25%. These results suggest that MFEFM enhances multiscale representation and models fuzzy boundaries more effectively, thereby reducing false alarms caused by insufficient low-level semantics and missed detections caused by high-level detail loss.

Finally, FEENet, which integrates DFFM, EAM, and MFEFM simultaneously, achieves the best results on all datasets. Compared with the baseline, the OA increases by 0.17% and 0.26%, the mIoU increases by 1.45% and 2.41%, and the mF1 increases by 0.89% and 1.54%. These results highlight the complementary strengths of the three modules; the fine-grained edge features extracted by EAM are effectively injected into the multiscale representations refined by MFEFM, while DFFM provides reliable temporal difference cues. This joint design leads to substantial performance improvements in challenging CD scenarios.

3) Selection of β: Ablation studies were conducted on the LEVIR-CD dataset to examine the effect of β on the model's performance, utilizing β values from 0.2 to 0.6 and evaluating with OA, mIoU, and mF1 metrics. The results are presented in Table VIII. Overall, the model performs well within this range, achieving optimal performance at $\beta = 0.4$, where OA reaches 99.18%, mIoU 92.01%, and mF1 95.69%. The parameter β acts as a crucial balancing factor in the residual connection of the EAM, governing the contribution of the frequency-domain enhanced features to the original low-level features. The core tradeoff lies between reinforcing genuine change-related edges

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TABLE VIII
ABLATION STUDY ON β

β values	Accurary		
	OA(%)	mIoU(%)	mF1(%)
β = **0.2**	99.15	91.82	95.58
β = **0.3**	99.16	91.87	95.61
β = **0.4**	**99.18**	**92.01**	**95.69**
β = **0.5**	99.15	91.86	95.61
β = **0.6**	99.15	91.80	95.57

TABLE IX
PERFORMANCES OF ZERO SHOT FOR VARIOUS METHODS

Methods	Levir-CD → WHU-CD		
	OA(%)	mF1(%)	mIoU(%)
FC-EF	87.94	65.34	55.22
FC-Diff	92.92	73.14	63.03
FC-Conc	94.24	76.26	66.27
BIT	96.97	82.62	73.53
ChangeFormer	97.80	84.20	75.52
DMINet	97.62	83.07	74.18
CF-GCN	96.99	77.88	68.55
SAM-CD	98.58	91.49	85.26
Meta-CD	**98.70**	**91.65**	**85.41**
Ours	98.61	91.51	85.27

and amplifying high-frequency noise. A lower β provides insufficient emphasis on critical edge structures, limiting the module's ability to enhance subtle boundaries. Conversely, a higher β overamplifies the high-frequency components, which can disproportionately accentuate imaging noise and irrelevant texture details, thereby introducing instability and ambiguity into the edge features provided to subsequent modules. The results demonstrate that $\beta = 0.4$ effectively balances this tradeoff, optimizing the fusion weight of high-frequency edge information to enhance the representation of salient edge details while maintaining robustness against noise. This value ensures that the EAM provides strong, yet clean, edge guidance to the MFEFM, facilitating effective collaboration between the two modules. Deviations from this value, either higher or lower, disrupt this balance and lead to reduced CD accuracy. Therefore, we set $\beta = 0.4$ as the default value in the FEENet.

F. Capability of Zero-Shot Transfer

In the RSCD task, the characteristics and imaging styles of different datasets vary considerably, which often requires traditional CD models to be retrained for each dataset. The LEVIR-CD dataset mainly focuses on building changes in urban areas, whereas the WHU-CD dataset contains a mixture of urban and rural scenes. Since conventional RSCD methods are typically optimized for the distribution of the training data, their performance often drops substantially when they are applied to datasets with different styles. To evaluate cross-dataset generalization, we trained FEENet exclusively on the LEVIR-CD dataset and tested it directly on the WHU-CD dataset. The results are shown in Table IX. Traditional CD approaches, such as FC-EF, FC-Diff, FC-Conc, BIT, ChangeFormer, DMINet, and CF-GCN, exhibited a significant decline in accuracy when transferred to WHU-CD. These models generally produced many incorrect predictions, and their best performance, achieved by ChangeFormer, reached only 97.80% in OA, 84.20% in mF1, and 75.52% in mIoU. In comparison, FEENet obtained 98.61% in OA, 91.51% in mF1, and 85.27% in mIoU. These results clearly show that FEENet surpasses all traditional baselines across all three evaluation metrics. Although its performance is slightly lower than Meta-CD, FEENet maintains a level of zero-shot generalization ability that is consistent with other methods based on FastSAM, remaining stable across datasets that differ in scene composition and imaging conditions. These experimental results verify that our FastSAM-based model preserves strong zero-shot transferability, allowing it to maintain robust performance even when evaluated on previously unseen datasets.

G. Discussion

The satisfying performance of FEENet can be attributed to its novel integration of the EAM and the MFEFM. The design of the EAM enables the network to enhance structural details in the frequency domain and suppress background noise, which significantly improves the accuracy of edge localization. By incorporating the MFEFM, FEENet dynamically injects refined edge information and models boundary uncertainty with Gaussian fuzzy logic, enabling the model to better detect subtle changes in complex regions. However, despite these advantages, FEENet has some limitations. First, its effectiveness is inherently tied to the powerful but generic feature representation of the pretrained FastSAM, which may constrain specialization on specific RS domains. Second, although based on the efficient FastSAM, the introduction of the additional ResNet-18 branch and the sophisticated EAM and MFEFM results in higher computational complexity and inference time compared to other lightweight CD methods that also utilize FastSAM.

V. CONCLUSION

Current CD algorithms based on FastSAM exhibit notable limitations in edge detection. To address this issue, this article proposes an edge-enhanced CD framework named FEENet, built upon FastSAM. The framework employs a collaborative encoder combining ResNet-18 and FastSAM for feature extraction and uses the extracted high-level semantic features to guide and focus the learning process on the changed regions. An EAM is designed to accurately extract edge details through the fusion of frequency-domain enhancement and semantic features. Furthermore, an MFEFM is innovatively designed, which utilizes Gaussian fuzzy logic to model edge uncertainty and dynamically injects refined edge information into multilevel change features, thereby effectively strengthening the correlation between edge semantics and change regions. Through comprehensive evaluation on the five public datasets LEVIR-CD, WHU-CD, SYSU-CD, CLCD, and S2Looking, FEENet is shown to outperform current SOTA approaches, particularly in detecting subtle changes in edge regions such as building contours and road boundaries. These results confirm

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the framework's capability and robustness in addressing edge-related challenges in VHR RSI CD tasks.

REFERENCES

[1] G. Cheng et al., "Change detection methods for remote sensing in the last decade: A comprehensive review," *Remote Sens.*, vol. 16, no. 13, p. 2355, Jun. 2024.

[2] P. Coppin, I. Jonckheere, K. Nackaerts, B. Muys, and E. Lambin, "Review ArticleDigital change detection methods in ecosystem monitoring: A review," *Int. J. Remote Sens.*, vol. 25, no. 9, pp. 1565–1596, May 2004.

[3] H. Wang, Y. Liu, Y. Wang, Y. Yao, and C. Wang, "Land cover change in global drylands: A review," *Sci. Total Environ.*, vol. 863, Mar. 2023, Art. no. 160943.

[4] D. Qin et al., "MSIM: A change detection framework for damage assessment in natural disasters," *Expert Syst. Appl.*, vol. 97, pp. 372–383, May 2018.

[5] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[6] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

[8] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[9] W. Zhang, W. Guo, Y. Li, and W. Xie, "Change detection meets frequency learning: A coarse-to-fine dual-domain detection network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5618513.

[10] T. Zhu et al., "FTA-Net: Frequency-temporal-aware network for remote sensing change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 3448–3460, 2025.

[11] Y. Xing, Y. Jia, S. Gao, J. Hu, and R. Huang, "Frequency-enhanced mamba for remote sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[12] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[13] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[14] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[15] T. Chen, Z. Lu, Y. Yang, Y. Zhang, B. Du, and A. Plaza, "A Siamese network based U-Net for change detection in high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2357–2369, 2022.

[16] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet+: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Jun. 2020.

[17] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409818.

[18] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[19] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[20] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[21] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[22] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[23] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[24] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.

[25] C. Lu, X. Zhang, K. Du, H. Xu, and G. Liu, "CTCFNet: CNN-transformer complementary and fusion network for high-resolution remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5408717.

[26] W. Xie, W. Shao, D. Li, Y. Li, and L. Fang, "MIFNet: Multi-scale interaction fusion network for remote sensing image change detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 3, pp. 2725–2739, Mar. 2025.

[27] W. Wang, C. Liu, G. Liu, and X. Wang, "CF-GCN: Graph convolutional network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607013.

[28] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[29] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.

[30] J. Gao, D. Zhang, F. Wang, L. Ning, Z. Zhao, and X. Li, "Combining SAM with limited data for change detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614311.

[31] X. Wang, X. Zhang, Y. Cao, W. Wang, C. Shen, and T. Huang, "SegGPT: Towards segmenting everything in context," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 1130–1140.

[32] X. Zou et al., "Segment everything everywhere all at once," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 19769–19782.

[33] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8778–8787.

[34] W. Guan, T. Wang, J. Qi, L. Zhang, and H. Lu, "Edge-aware convolution neural network based salient object detection," *IEEE Signal Process. Lett.*, vol. 26, no. 1, pp. 114–118, Jan. 2019.

[35] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

[36] B. Bai, W. Fu, T. Lu, and S. Li, "Edge-guided recurrent convolutional neural network for multitemporal remote sensing image building change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5610613.

[37] Z. Chen et al., "EGDE-Net: A building change detection method for high-resolution remote sensing imagery based on edge guidance and differential enhancement," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 203–222, Sep. 2022.

[38] L. Xia, J. Chen, J. Luo, J. Zhang, D. Yang, and Z. Shen, "Building change detection based on an edge-guided convolutional neural network combined with a transformer," *Remote Sens.*, vol. 14, no. 18, p. 4524, Sep. 2022.

[39] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015.

[40] Y. Zhu, K. Lv, Y. Yu, and W. Xu, "Edge-guided parallel network for VHR remote sensing image change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 7791–7803, 2023.

[41] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11531–11539.

[42] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[43] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4403718.

[44] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[45] L. Shen et al., "S2Looking: A satellite side-looking dataset for building change detection," *Remote Sens.*, vol. 13, no. 24, p. 5094, Dec. 2021.

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[46] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 8026–8037.
[47] K. Li et al., "Open-CD: A comprehensive toolbox for change detection," 2024, *arXiv:2407.15317*.



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